

Known as the creator of the periodic table, Dmitri Mendeleev was an eccentric chemistry professor who solved the problem of how the elements are related to each other. His classification system advanced science, leading to the discovery of new elements and an understanding of the structure of the atom.

MILESTONES

CHEMISTRY PIONEER

Discusses standardizing chemistry at the first-ever international chemistry conference in 1860.

PERIODIC TABLE

His brainwave comes in 1869, when he devises his periodic table of elements in a grid pattern.

FIRST REFINERY

Conducts research into petroleum and helps to found the first Russian oil refinery in 1879.

WINS ACCLAIM

In 1905, is awarded the Copley Medal and elected to the Royal Swedish Academy of Sciences.

PRIZE NOMINEE

Nominated for the Nobel Prize in Chemistry in 1906, but is rejected, possibly due to a personal grudge.

Dmitri Ivanovich Mendeleev was born into a poor Russian family in a village in Siberia, the youngest of 14 surviving children. When Mendeleev was 13, his father went blind, and his mother had to support the family by running a glass business. Two years later, the factory burned down. Determined that Mendeleev should have a higher education, his mother walked and hitchhiked with him across Russia to St. Petersburg. Tragically, she died just days after he enrolled at the university there.

Mendeleev shone at his studies and later became professor of chemistry. He was an idiosyncratic teacher, known for his disheveled appearance and fiery temper. His career flourished, however, and his textbook *The Principles of Chemistry* was published between 1868 and 1870; it has since been translated into several languages.

One of the key challenges within chemistry at the time was how to classify the elements. Several scientists had identified certain patterns in the way elements behaved—for example, some reacted violently with others, while others hardly seemed to react at all. It was known that

Mendeleev wrestled

with the puzzle of how all of the elements fit together. Inspiration finally came to him one day after he fell asleep at his desk.

